



PH AND ITS CLINICAL SIGNIFICANCE IN DENTISTRY

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OBJECTIVES:

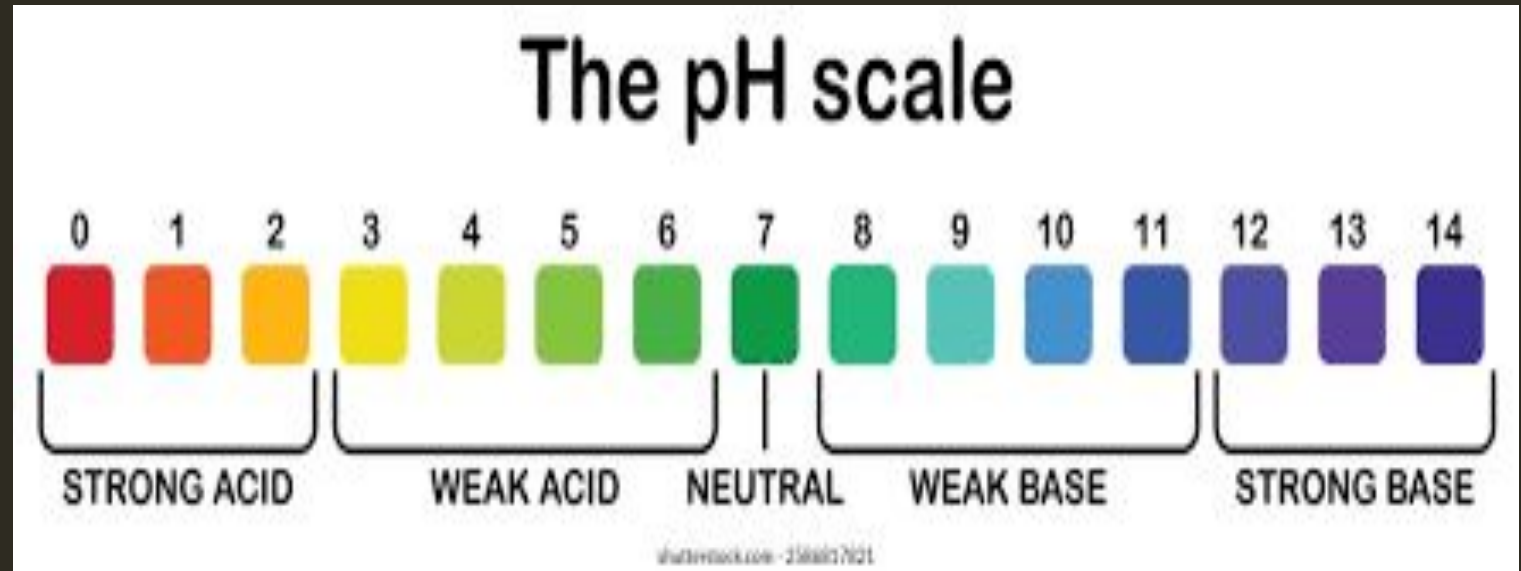
- Definition of pH
- Importance in biological systems
- Relevance to oral health and dentistry

DEFINITION OF PH

- pH = potential (power) of hydrogen
- Measure of hydrogen ion concentration $[H^+]$
- Formula: $pH = -\log_{10}[H^+]$
- Sorensen introduced the concept in 1909

PH SCALE

- Range: 0 – 14
- Acidic: <7
- Neutral: 7
- Alkaline: >7



- Logarithmic scale (each unit = 10x change in $[H^+]$)

STRONG VS WEAK ACIDS

- Strong acids: Complete dissociation
- Weak acids: Partial dissociation
- Importance in biological systems
- Examples:
 - HCl (strong acid)
 - Carbonic acid (weak acid)

IONIZATION OF WATER

- Water dissociation:



- Ionic product of water (K_w)

$$K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$$

- At neutral pH: $[\text{H}^+] = [\text{OH}^-] = 10^{-7} \text{ M}$

HENDERSON-HASSELBALCH EQUATION

- $\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$
- Used to calculate pH of buffer solutions
- Important in blood and saliva buffering

BUFFERS

- Definition: Solutions that resist change in pH upon addition of small amounts of acid or base.
- Components:
 - Weak acid
 - Conjugate base (salt)

NORMAL PH OF BODY FLUIDS

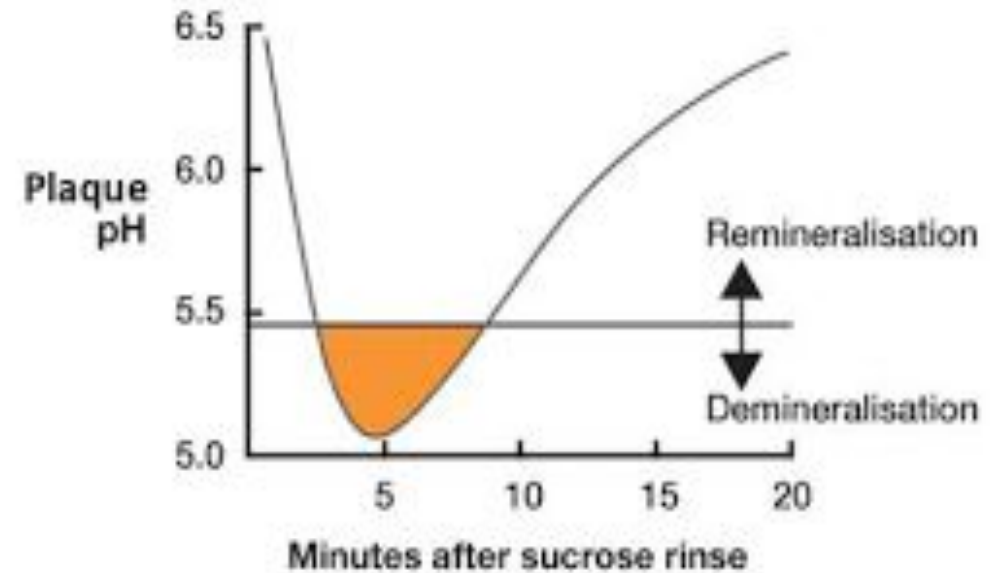
- Blood: 7.35 - 7.45
- Saliva: 6.2 - 7.6
- Gastric juice: 1.5 - 3.5
- Urine: 4.5 - 8

PH AND DENTAL CARIES

- Critical pH for enamel: 5.5
- Below this → enamel demineralization
- Caused by acids produced by oral bacteria
- e.g., *Streptococcus mutans*

STEPHAN CURVE

- Shows drop in plaque pH after sugar intake
- Rapid fall below critical pH
- Gradual recovery due to saliva buffering



ROLE OF SALIVA IN PH CONTROL

- Contains bicarbonate buffer
- Washes away acids
- Maintains remineralization

Clinical importance in xerostomia patients.

SUMMARY

- pH measures hydrogen ion concentration
- Logarithmic scale
- Buffers maintain physiological pH
- Critical pH (5.5) important in dentistry
- Saliva plays protective role

